

DSAA3073

Course Overview & Logistics

Prof. Wei Wang

HKUST(GZ)

2026-06-15

Outline

Course Logistics

New Learning Method

Using the Learning Sheet

Using AI Tools

Course Content Overview

Summary

Schedule & Location

Lecture (Assisted self-paced learning)

- Mon/Wed, 1030–1320
- W1-222

Tutorial (Group-based self-paced learning, optional)

- Mon/Wed, 1800–1850
- W1-222

Notes

- This course uses an **experimental** learning method that requires in-class attendance, learning, and use of AI/LLM.
 - Drop period is 17th June 2026, so **remember** to drop it if you are not comfortable with the method.
 - If you can survive, you may thrive in **the AI era** — the method has been tried in DSAA3071 26Spring with promising results: majority of students think they can learn “deeper”, many reported “deeper and faster”.
- Bring a laptop/tablet for lectures and tutorials.

Teaching Team

Instructor

Prof. Wei Wang
weiwcs@ust.hk

Tutor

Duyi PAN
dpan457@connect.hkust-gz.edu.cn

Assessment Structure

Weekly Tests

- 12 validation tests
- 30 minutes each
- Total: **70%**

Assignment or Project

- Written or programming
- Total: **30%**

Key idea

Validation sheets and test sheets are **i.i.d.**: independent and identically distributed.

Practice sheets are direct test preparation.

Grading Scale

Grade	Min Score	Requirements
A+ A A-	85	Must be ranked in top 30%
B+ B B-	70	—
C+ C C-	60	—
D	50	—
F	< 50	—

Subgrades (A-, B+, etc.) assigned based on overall distribution.

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Why This Course Is Worth the Effort

According to **Expectancy-Value Theory (EVT)**, effort rises when value is high, success feels attainable, and cost is acceptable.

Utility value

- Foundational for ML/AI research and practice.
- Bridges theory and application in data science.

Intrinsic value

- Keep curiosity alive.
- Independent learning is a durable AI-era advantage.

Attainment value

- We provide learning sheets, tests, support, and AI tools.
- You convert them into competence and marks.

Cost

- Almost no extra cost: you must study anyway.
- In-class work can beat struggling alone; DSAA3071 feedback supports this.

Why This New Approach?

Pacing

- One lecture speed cannot fit everyone.
- Some are lost; others wait.

AI-era self-learning

- Practice attention, retrieval, and repair.
- Productive struggle is part of learning.

Reading beats passive listening

- Reading exposes gaps.
- Listening can create false mastery.

Use required time well

- You must learn anyway.
- Structured class time should make it faster.

How It Works: Session Format

1. 30-minute validation test

- Assessing previous week's material.

3. 150 minutes self-paced learning

- Learn at your pace.
- With AI + instructor support.

2. 10-minute break

- Recharge and reset.

4. Optional evening tutorials

- Group-based learning.
- Led by TA.

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Five Key Strategies

1. Follow the narrative

- Concepts build progressively.
- Do not jump around randomly.

2. Understand before memorizing

- Proofs show why techniques work.
- Do not memorize formulas in isolation.

3. Work through examples

- Examples and exercises teach.
- Do not treat them as decoration.

Five Key Strategies (continued)

4. Ask, and be ready to be asked

- I will circulate.
- Be ready to explain your thinking.

5. Ask for help when stuck

- Productive struggle is good.
- Spinning wheels is not.

Use all resources: materials, instructor, peers, and AI tools.

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AI Tool Requirements

Requirements

- Use the course AI setup.
- Use course-selected SOTA models.
- Avoid unrelated external AI tools in class.
- Usage is logged for attendance and analytics.

Purpose

- Use AI as a learning aid.
- Judge **when** and **how** to use it.
- Practice prompting and critique.
- Help improve the course.

How to Learn with AI Without Outsourcing Thinking

Read first

Interpret definitions, examples, and problem statements before asking AI.

Repair and transfer

Rewrite the solution yourself; test it on a nearby variant.

Ask for diagnosis

Prefer “where is my reasoning wrong?” to “give me the answer.”

Own the attainment

AI accelerates feedback; it cannot replace retrieval, judgment, or curiosity.

Optional resources

- Feynman Technique: explain simply; find gaps; repair; repeat.
- Socratic questioning: clarify assumptions, reasons, evidence, and alternatives.
- Intuition & mental models: build concrete pictures before formal symbols.
- How To Become Dangerously Self-Educated (with AI): <https://www.youtube.com/watch?v=VeU6gScy92s>

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Topics Covered

Tentative schedule (subject to adjustment based on class progress and feedback)

Weeks 1-5: Foundations

- Statistical learning theory
- Generalization bounds
- PAC learning framework
- VC dimension

Weeks 6-9: Algorithms

- Boosting (AdaBoost)
- Online learning
- Regret analysis
- Optimization methods

Weeks 10-12: Deep Learning Theory

- Neural network expressiveness
- Neural Tangent Kernel (NTK)
- Implicit regularization
- Generalization in overparameterized models

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Key Takeaways

Logistics

- Lectures: Mon/Wed 1030–1320
- Tutorials: Mon/Wed 1800–1850
- W1-222
- Bring laptop/tablet

Learning method

- Self-paced sheet work
- 12 weekly tests = 70%
- Assignment/project = 30%
- Active reading and repair
- Instructor, peers, and AI support

Use the new method to learn how to learn effectively.

Questions?

“一切教育本质上都是自我教育，一切学习本质上都是自学”

“Education is ultimately self-education, and learning is ultimately self-learning.”

— Guoping Zhou

Source: <https://www.sinoss.net/c/2016-03-09/524538.shtml>

Ready to start learning!